

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Atmospheric pollution: smoke, acid rain, NO_x, hydrocarbons, CO, smogs - Set 1

Mark scheme

AQA spec 3.4.3.2.1-3.4.3.2.5; Pollution; textbook pp. 245-254

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) (1) • Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) (1)	2
01.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(930 - 250) / 250 \times 100$ (1) • = 272% (1)	2
01.3	<i>1 mark for each point.</i> • Anticyclonic conditions: clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled (1) • Low wind velocity so air layers did not mix; London lies in a valley / basin where cold dense air collected: a temperature inversion formed (1) • Smoke from coal fires was trapped below the inversion; moist air cooled to its dew point forming fog; smoke + fog = smog; the smog's high albedo prolonged the inversion (1)	3
01.4	<i>1 mark for each point.</i> • Smoke particles (PM₁₀) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections (1) • These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal (1)	2
01.5	<i>1 mark.</i> • Clean Air Act (1956) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	

Question	Marking guidance	Marks
02.1	1 mark for each point. - Effluent gases pass through a chamber containing electrically charged wires / plates (1) - Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick (1) - Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials) (1)	3
02.2	1 mark for each point. - Effluent gases are forced to rotate / spin in a cylindrical chamber (1) - The (denser) particles are thrown outwards to the chamber walls where they fall and collect; cleaned gases leave through a central pipe (1)	2
02.3	1 mark. The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke (1)	1
02.4	1 mark for each point. - Turbochargers deliver more air / oxygen to the combustion chamber (1) - So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration) (1)	2
02.5	1 mark for each point. - Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled (1) - They penetrate deeper into the lungs / alveoli and are not cleared; may carry toxic / carcinogenic chemicals (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
03.1	1 mark for each point. - pH 5.6 (1) - Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid (1)	2
03.2	1 mark for each point. - Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) (1) - Granite does not neutralise acids so lake water in A is acidic (4.6) (1) - Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) (1) - Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost (1)	4
03.3	1 mark for each point. - Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines (1) - Dissolves / weakens limestone and marble buildings and statues (eg York Minster) (1)	2
03.4	1 mark for each point. - Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide (1) - Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Most sulfur in coal is present as (solid) iron pyrites (FeS_2) (1) • Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming) (1)	2
04.2	<i>1 mark for each point.</i> • Flue gases are passed through a bed of crushed calcium carbonate (limestone) (1) • Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) (1) • If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster (1)	3
04.3	<i>1 mark for each point.</i> • The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide (1) • The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource (1)	2
04.4	<i>1 mark for each point.</i> • Fluidised bed: increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NO_x (production prevented) (1) • Catalytic converter: platinum / palladium catalyst chemically reduces NO_x back to nitrogen and oxygen gases in the exhaust (1) • Urea sprayed into the hot flue / exhaust gases reacts with NO_x, converting it to nitrogen, carbon dioxide and water (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	
05.1	<i>1 mark for each point.</i> • Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening (1) • Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight (1) • NO_2 (from morning traffic) is broken down by UV to NO and monatomic oxygen; monatomic oxygen reacts with O_2 to form ozone (1) • Ozone lags behind the NO / NO_2 peaks (which are produced by the morning rush-hour traffic) because the reactions take time; falls at night when there is no sunlight (1)	4
05.2	<i>1 mark for each point.</i> • NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure (1) • NO is oxidised in the atmosphere (by oxygen / ozone) to NO_2, a secondary pollutant (1)	2
05.3	<i>1 mark for each point.</i> • PANs (peroxyacetyl nitrates) (1) • Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NO_x) (1)	2

Question	Marking guidance	Marks
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Heavy vehicle use giving high NO_x and hydrocarbon emissions (1) • Sunny climate providing the UV for photochemical reactions (1) • Valley / basin topography and frequent temperature inversions trapping pollutants (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
06.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Unburnt fuel in vehicle exhaust gases (1) • Evaporation from fuel tanks / spillage at filling stations (1) • Solvents (paints, cleaning products) (1) • Aerosol propellants (1) • Gas leaks / fossil fuel exploitation (methane) (1)	3
06.2	<i>1 mark for each point.</i> • Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank (1) • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon (1)	2
06.3	<i>1 mark for each point.</i> • A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe (1) • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) (1) • NO_x are chemically reduced to nitrogen and oxygen (1)	3
06.4	<i>1 mark for each point.</i> • Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues (1) • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Smoke: solid particulates (PM₁₀, PM₅, PM₁) from incomplete combustion; small particles stay suspended, penetrate lungs; toxic chemicals (acids, phenols, metals) carried on particles; high albedo cools climate; blocks light reducing photosynthesis; damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths); synergism with SO₂ • Controls: Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion; smoke smogs have largely been eliminated in UK cities • Acid rain gases: SO₂ (coal combustion, smelting sulfide ores) oxidised by ozone to SO₃ - sulfuric acid; NO_x (high temperature combustion) - nitric acid; HCl from PVC incineration; reactive gases travel long distances (transboundary pollution) • Effects: pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata); dissolves exoskeletons; leaches Ca and Mg then mobilises toxic Al and Pb; corrodes metals, dissolves limestone; severity depends on bedrock (limestone neutralises) • Controls: fuel desulfurisation (coal washing, hydrosulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays; switching from coal to gas / renewables • Evaluation: SO₂ emissions in Europe have fallen dramatically and lake pH is recovering; NO_x harder to control because of growing vehicle numbers; catalytic converters need lead-free fuel and warm engines; FGD is costly and produces wastes / gypsum; controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO₂ vs NO_x	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Smoke is particulate matter from incomplete combustion of coal, diesel, wood and wastes. Its effects follow from particle size: PM10 and smaller particles stay suspended, travel far and penetrate the lungs, carrying acids, phenols and metals; smoke also reflects sunlight (cooling), blocks light for photosynthesis and blackens buildings. Under a temperature inversion in a valley it accumulates as smog, and it acts synergistically with sulfur dioxide - the London smog of 1952 killed about 12 000 people. Acid rain gases are different: sulfur dioxide from coal combustion and sulfide smelting, and nitrogen oxides from high-temperature combustion, are reactive and soluble, forming sulfuric and nitric acids that travel long distances across borders. Low pH denatures proteins in exposed cells such as gills, eggs and root hairs, dissolves exoskeletons and limestone, corrodes metals and leaches calcium and magnesium from soils while mobilising toxic aluminium, with severity depending on bedrock.	
	Smoke control has been the clearest success: the Clean Air Act's smokeless zones, the switch to gas and electricity, electrostatic precipitators, cyclones, bag filters and diesel particulate filters have virtually eliminated smoke smogs in developed countries. Sulfur dioxide has also fallen dramatically through fuel desulfurisation, flue-gas desulfurisation and the move from coal to gas and renewables, and lake pH is recovering, although FGD is costly and produces waste. Nitrogen oxides remain the hardest: catalytic converters and fluidised-bed combustion work, but converters need unleaded fuel and warm engines and vehicle numbers keep rising. Controls have therefore succeeded for particulates and sulfur, only partly for nitrogen.	
	Weaker answer - Level 2 (5 marks)	
	Smoke is made of small particles from burning coal and diesel. It causes breathing problems and smogs like in London in 1952. Acid rain is caused by sulfur dioxide and nitrogen oxides which dissolve in rain and make it acidic. Acid rain kills fish and trees and damages buildings. Smoke is controlled by smokeless zones and filters. Acid rain is controlled by taking the sulfur out of fuel and catalytic converters in cars. These methods have reduced pollution a lot, especially smoke.	
	What makes the difference	
	• Level 3 links each effect to a property (particle size, solubility, reactivity, pH denaturing proteins); Level 2 describes effects without the mechanism.	
	• Level 3 evaluates each control separately with evidence (recovering lake pH, cost of FGD, why NO _x is harder); Level 2 gives one general sentence.	
	• Both answers use the 1952 smog, but Level 3 explains the conditions (inversion, synergism with SO ₂) that caused it.	
07.2	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Reduces growth rates of trees / crops (1) • Damages leaf cuticles (allowing SO ₂ to cause more damage) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	
08.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(6400 - 220) / 6400 \times 100$ (1) • = 96.6% (accept 97%) (1)	2

Question	Marking guidance	Marks
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO₂ source (1) • Flue gas desulfurisation fitted to remaining coal plants; coal / oil desulfurisation (1) • NO_x comes mainly from road transport and vehicle numbers / distance travelled increased greatly (1) • NO_x is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur (1) • Catalytic converters only work when warm / diesel engines produce more NO_x / older vehicles remained in use (1)	3
08.3	<i>1 mark for each point.</i> • Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO₂ (broken down by UV to monatomic oxygen which forms ozone) (1) • If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form (1)	2
08.4	<i>1 mark for each point.</i> • Lichens are very sensitive to acidic conditions / SO₂ and different species have different tolerances (1) • The species present, their abundance, size / health can be used as a biotic index of the pollution level (1) • They give a long-term integrated measure of pollution rather than a single instantaneous reading; cheap, no equipment needed (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	